
Globally Optimal Nonconvex Optimization of Smoothed Hamilton-Jacobi PDEs for Scaled Reachability

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Abstract

1 Computing safe reachable sets in verification settings for high-dimensional dy-
2 namic systems, parameterized by the viscous Hamilton-Jacobi (HJ) equation, in
3 general involves grid discretization of the phase space to achieve a numerically
4 consistent solution with respect to the original non-viscous problem. This presents
5 a two-fold problem: discretization implies exponential memory requirements; and,
6 computational costs increase as problem dimensions multiply. While parallel com-
7 puting can reduce the impact of the latter, it does not address the grid discretization
8 exponential memory costs of the former. Leveraging a recent connection between
9 HJ value functions, proximity operators, and the Moreau envelopes of HJ PDEs,
10 we cast the recovery of the safe set as an adaptive gradient descent optimization of
11 the Moreau envelope of the reachable HJI variational PDE. Our scheme provides
12 a storage-free means for resolving safe sets in a computationally time-efficient
13 manner without dealing with the exponential memory costs of grid discretization.

14 1 Introduction

15 This paper takes a further step towards improved computational and scalable solutions to time-
16 dependent Hamilton-Jacobi (HJ) partial differential equations (PDEs) (author?) (Eva22) employed
17 in the safety verification (author?) (HMA⁺23) of dynamical systems. We work within a viscous HJ
18 reachability ramification (author?) (Lyg04, MBT05, lev24, Mol25) and establish our results with
19 globally optimal solutions to reachability problems in nonlinear settings. In emphasis, we consider
20 first-order nonlinear scalar Hamilton-Jacobi (HJ) PDEs of the form

$$v_t + \mathbf{H}(t; \mathbf{x}, \partial_x v) = 0 \text{ in } \mathbb{R}^n \times (0, T], \quad v(0; \mathbf{x}) = \mathbf{g}(\mathbf{x}) \text{ on } \mathbb{R}^n \times \{t = 0\} \quad (\text{HJ-IVP})$$

21 where v_t denotes the partial derivative of the vector field $v(\mathbf{x}, t)$ with respect to t ; the Hamiltonian
22 $\mathbf{H}(\mathbf{x}, t)$ is continuously defined on $\mathbb{R}^n$; $\mathbf{g}(\mathbf{x})$ is bounded and uniformly continuous (BUC) in $\mathbb{R}^n$;
23 and $\partial_x v$ is the spatial gradient of $v(\mathbf{x}, t)$.

24 While a unique C^2 solution $v(\mathbf{x}, t)$ exists uniformly on $0 \leq t < T$ to (HJ-IVP) for which
25 $\lim_{t \rightarrow T} v(\mathbf{x}, t)$ (when $\mathbf{H}$ and $\mathbf{g}$ are smooth and $v_0(\mathbf{x}, t)$ is compactly supported), $\mathbf{H}(\mathbf{x}, t)$ lacks
26 continuous derivatives in this limit so that $\partial_x v$ vanishes at $t = T$. However, one may find Lipschitz
27 continuous functions $v(\mathbf{x}, t)$ on $\mathbb{R}^n \times (0, T]$ which are smooth *almost everywhere* (a.e.) and satisfy
28 (HJ-IVP) in a “generalized sense”. Suppose that we fix a positive $\delta > 0$, Crandall and Lions (author?)
29 (CL84) showed that if $\delta > 0$, then v^δ is the *vanishing viscosity* solution of

$$v_t^\delta + \mathbf{H}(t; \mathbf{x}, \partial_x v^\delta) = \frac{\delta}{2} \Delta v^\delta \text{ in } \mathbb{R}^n \times (0, T], \quad v^\delta(0; \mathbf{x}) = \mathbf{g}(\mathbf{x}) \text{ on } \mathbb{R}^n \times \{t = 0\} \quad (\text{HJ-Viscous})$$

with Laplacian Δv^δ and a constant $k > 0$. Uniform convergence of $v^\delta \rightarrow v$ exists in the limit $\delta \rightarrow 0^+$, i.e.

$$\sup_{t \in (0, T)} \sup_{x \in \mathbb{R}^n} |v(t; x) - v^\delta(t; x)| \leq k\sqrt{\delta}. \quad (1)$$

Equation (HJ-Viscous) models conservation laws in a single spatial dimension (author?) (OS91). For inf-sup or sup-inf optimal control problems (author?) (Lyg04), the Hamiltonian is related to the *backward* reachable set (author?) (MBT05) of a dynamical system. Reachable systems are those where one can compute all states that can be reached from an initial condition in a *finite number of steps*.

Popular numerically consistent optimal solutions (author?) (CL84, CM80) to (HJ-Viscous) utilize problem space grid discretization (author?) (OF04, MBT05). However, these exact an exponential memory storage — making their practical applications on large problems limited. Our recent efforts (author?) (lev24, Mol25) leveraged single-instruction, multiple threads (SIMT) with modern GPUs to accelerate computation on a host of practical problems; we demonstrated improved spatial scalability with up to 76 – 92% improvement in computation for higher-dimensional problems. However, it was grid-based and as problem dimensions increase, the need for more memory may make such approaches unattractive to users. We thus tack another thread, jettisoning grid-discretization methods for sampling-based proximal operators in a statistical moving average context. This resolves the global extrema of the Moreau envelopes of HJ cost functions, which can be used to recover the reachable set in nonconvex optimization settings (author?) (HWFO24, CL23, COO⁺18, CCS⁺19).

The Cole-Hopf transformation (author?) (Eva22, COO⁺18) employed in PDE analysis relates the heat equation solution to that of the viscous HJ equation’s solution. Recently, Osher et. al (author?) (OS91) applied the Cole-Hopf transformation to v^δ and related the resulting transformed heat conservation equation to an importance sampling scheme for approximating the solution v^δ of (HJ-Viscous). We build on this, introducing a stochastic and adaptive Moreau gradient descent scheme for optimizing the value function under two inputs in a differential game setting. Ours is a sampling scheme that sidesteps essentially nonoscillatory (author?) (OS91) finite-differencing Lax-Friedrichs schemes (author?) (OF04) in favor of a gradient descent optimization on the Moreau envelope of the value function. This allows us to resolve *robustly controlled backward reachable sets and tubes* (author?) (Mit20) in the terminal Hamilton-Jacobi-Isaacs equation (author?) (Isa99, MBT05, lev24).

The rest of this paper is structured as follows: in §2, we provide the theoretical machinery for our problem. In §3, we describe our methods. Section 4 presents our numerical algorithms on sample problems and we conclude the paper in §5.

2 Background

We first introduce the notations that are commonly throughout this article. Reachable sets within the context of two person games (author?) (Isa99) and their accompanying “viscous” terminal HJI PDE (author?) (ES84a) are then introduced. This is followed by the HJ-Isaacs PDEs commonly used to characterize reachable sets.

2.1 Notations and Terminologies

Conventions: Upper-case and lower-case bold font Roman letters are matrices and vectors, respectively; while calligraphic letters are sets. Exception: the subdifferential *set* of a BUC function, $g(x)$, is denoted ϑ . Time variables e.g. t_0, t, τ, T will always be real positive numbers. The n -dimensional Euclidean space is $\mathbb{R}^n$; the set of natural numbers is $\mathbb{N}$. The natural log of T is denoted $\log T$. The dot product is $\langle \cdot, \cdot \rangle$. The partial time derivative of v is $\partial_t v$ while its partial spatial derivative is denoted $\partial_x v$. For a function $f(x(t), y(t))$ with every argument parameterized by t , we simply write $f(t; x, y)$. We represent the real-valued Hilbert space with $\mathcal{H}$; denote $2^{\mathcal{H}}$ by its power set, i.e. the family of all subsets of $\mathcal{H}$. Associated with $\mathcal{H}$ is the maximally monotone set-valued operator, $Q : \mathcal{H} \rightarrow 2^{\mathcal{H}}$. The zeroes of Q are denoted $\text{zer } Q = \{x \in \mathcal{H} \mid 0 \in Qx\}$. The set of global minimizers of $g(x)$ is denoted $\text{Argmin } g(x)$. It is written as $\text{argmin } g(x)$ when the set is a singleton. The set of fixed point of operator $S : \mathcal{H} \rightarrow \mathcal{H}$ is $\text{Fix } S = \{x \in \mathcal{H} \mid Sx = x\}$.

79 The unique solution $\mathbf{x} \in \mathbb{R}^n$ of the dynamical system $\dot{\mathbf{x}}(\tau) = f(t; \mathbf{x}, \mathbf{u}, \mathbf{w})$, influenced by a pursuing
80 player $\mathbf{P}$ (with control $\mathbf{w} \in \mathcal{W} \subseteq \mathbb{R}^p$) and its evading pair $\mathbf{E}$ (with control $\mathbf{u} \in \mathcal{U} \subseteq \mathbb{R}^m$), results in
81 system trajectories $\xi_{\mathbf{x}, \tau}^{\mathbf{u}, \mathbf{w}}$ that are obtained by optimizing a scalar-valued objective

$$\mathcal{J}(\tau; \mathbf{x}, \mathbf{u}, \mathbf{w}) = \min_{\tau \in [t, T]} g(\xi_{\mathbf{x}, \tau}^{\mathbf{u}, \mathbf{w}}(\tau)) \quad (\text{Target set cost})$$

82 within a two-person differential game for a fixed $T \geq \tau > t$. The controllers belong in compact sets
83 which are measurable functions i.e.

$$\bar{\mathcal{U}} \equiv \mathbf{u} : [t, T] \rightarrow \mathcal{U}, \quad \bar{\mathcal{W}} \equiv \mathbf{w} : [t, T] \rightarrow \mathcal{W}. \quad (2)$$

84 2.2 Lower value of the differential game

85 For a (possibly infinite) *terminal time* T , a constant k , $\mathbf{u} \in \mathcal{U}$, $\mathbf{w} \in \mathcal{W}$ and all $\hat{\mathbf{x}}$, $\mathbf{x} \in \mathbb{R}^n$, let the
86 system possess a BUC payoff functional $g(\mathbf{x}(T))$ that satisfies $g : \mathbb{R}^n \rightarrow \mathbb{R}$. The differential game's
87 *lower value* (**author?**) (ES84b)

$$v^-(\mathbf{x}, t) = \inf_{\beta \in \mathcal{B}(t)} \sup_{\mathbf{u} \in \mathcal{U}(t)} \mathcal{J}(t; \mathbf{x}, \mathbf{u}, \mathbf{w}) \triangleq \inf_{\beta \in \mathcal{B}(t)} \sup_{\mathbf{u} \in \mathcal{U}(t)} \min_{\tau \in [t, T]} g(\xi_{\mathbf{x}, \tau}^{\mathbf{u}, \mathbf{w}}(\tau)). \quad (3)$$

88 is the viscosity solution to (HJ-IVP) (**author?**) (ES84b, Lemma 2.1) with Hamiltonian,

$$H^-(t, \mathbf{x}, \mathbf{u}, \mathbf{w}, p) = \max_{\mathbf{u} \in \mathcal{U}} \min_{\mathbf{w} \in \mathcal{W}} \langle f(t, \mathbf{x}, \mathbf{u}, \mathbf{w}), p \rangle. \quad (4)$$

89 Notice in (3) that the pursuer possesses a non-anticipative strategy $\mathcal{B}(t)$ that is aware of all (maximiz-
90 ing) controls of the evading player up until time t . In what follows, we drop the negative superscript
91 to refer to the value of the game in the context of reachability analysis.

92 2.3 Robustly Controlled Backward Reachable Set and Tube

93 Suppose that the goal of $\mathbf{E}$ is to reach a user-specified target region $\mathcal{L}_0(\tau)$ within $\tau \leq T$ time steps
94 of playing the game; while the $\mathbf{P}$ simultaneously seeks to prevent this from happening. The invariant
95 set $\mathcal{L}_0$ obtained at T

$$\mathcal{L}_0(T) = \{\mathbf{x} \in \mathbb{R}^n \mid g(0; \mathbf{x}) \leq 0\} \quad (\text{Target-Set})$$

96 is “*robustly controlled*” for the “*distance-to-target-set*” cost $g(0; \mathbf{x})$, with constraints $|g(0; \mathbf{x})| \leq$
97 k , $|g(0; \mathbf{x}) - g(t; \hat{\mathbf{x}})| \leq k|\mathbf{x} - \hat{\mathbf{x}}|$ where $k > 0$ is a constant, and all $-T \leq t \leq 0$, $\{\hat{\mathbf{x}}, \mathbf{x} \in \mathbb{R}^n\}^1$.
98 The distance to $\mathcal{L}_0(\tau)$ is typically found by optimizing $g(\mathbf{x}, t)$ as in (Target set cost). The HJI PDE
99 (3) and Hamiltonian (4) for this set admit the form

$$v_t(t; \mathbf{x}) + \min\{0, H(\mathbf{x}, \partial v_{\mathbf{x}}(t; \mathbf{x}))\} = 0, \quad v(0; \mathbf{x}) = g(0; \mathbf{x}). \quad (\text{HJI-RCBRT})$$

100 For the safety problem setup in (3), the corresponding *robustly controlled backward reachable tube*
101 (RCBRT) on $(0, T]$ is the closure of the open set (**author?**) (Mit20)

$$\mathcal{L}([\tau, 0], \mathcal{L}_0) = \{\mathbf{x} \in \mathbb{R}^n \mid \exists \beta \in \bar{\mathcal{W}}(t) \forall \mathbf{u} \in \mathcal{U}(t), \exists \tau \in [-T, 0], \xi(\bar{t}) \in \mathcal{L}_0\}. \quad (\text{Target-Tube})$$

102 Read: The set of states from which the strategies of $\mathbf{P}$ and for all controls of $\mathbf{E}$ imply that we *reach*
103 *and remain in the target set* in the interval $[-T, 0]$. Following Lemma 2 of (**author?**) (MBT05), the
104 states in the reachable set admit the following properties w.r.t the value function v^- :

$$\mathbf{x}(t) \in \mathcal{L}(\cdot) \implies v(\mathbf{x}, t) \leq 0, \quad v(\mathbf{x}, t) \leq 0 \implies \mathbf{x}(t) \in \mathcal{L}(\cdot). \quad (\text{Zero Levelset})$$

105 In backward reach avoid tubes where the agent must avoid the unsafe region at all times, we can
106 write (HJI-RCBRT) as the HJI variational inequality for a *robustly controlled backward reach-avoid*
107 *tube* (RCBRAT) as follows,

$$\min\{v_t(t; \mathbf{x}) + H(t; \mathbf{x}, \partial v_{\mathbf{x}}), g(\mathbf{x}, t) - v(t; \mathbf{x})\} \leq 0, \quad v(0; \mathbf{x}) = g(0; \mathbf{x}). \quad (\text{HJI-RCBRAT})$$

108 **Challenges in optimizing the RCBRT/RCBRAT:** Computing the RCBRT presents a host of
109 challenges for optimization because

¹Time is reversed in BRT computational scenarios.

- the dynamics $f(\cdot)$ and the nonlinear PDE are usually nonlinear with coupled states in most problems of interest;
- although $\mathcal{L}([\tau, 0], \mathcal{L}_0)$ is mostly smooth, it possesses discontinuous “shocks” at solution crossover points (see (author?) (Mit20)); this makes global gradient-based optimization impossible for resolving $\mathbf{v}$;
- more than one trajectory can emanate from the RCBRT’s boundary that end up at the same point on the target set – this makes sampling-based optimization difficult, but not insurmountable; and
- the RCBRT is nonconvex.

To mitigate these challenges above, we introduce a set of tools that allow us to circumvent these challenges in an optimization context.

3 Methods

We first transform the nonlinear hyperbolic “smoothed” HJ PDE (HJ-Viscous) to a linearized form and exact its unique solution based on a deduction from the solution to the heat equation. We then prescribe a sampling-based scheme, based on proximity theory, for its spatial gradients. This is found in a stochastic gradient descent algorithm in a zeroth-order optimization HJ scheme. The latter parts of this section extend the results to (HJI-RCBRT) and (HJI-RCBRAT).

3.1 HJ Linearization by a Cole-Hopf Transformation

Here, we first transform the smooth HJ equation (HJ-Viscous) into a linear form via the Cole-Hopf trick (author?) (Eva22, §4.4.1). The Cole-Hopf transformation that linearizes (HJ-Viscous) can be written as (see the derivation in Appendix A)

$$\varphi^\delta = \exp(-2\mathbf{v}^\delta/\delta). \quad (\text{Cole-Hopf})$$

Applying the Cole-Hopf transformation to (HJ-Viscous), we find that the fundamental solution to the smoothed version of the Hamilton-Jacobi equation (HJ-Viscous) is given by (5).

Lemma 1 (Smoothed HJ Solution). *The unique solution to the smoothed version of the Hamilton-Jacobi equation (HJ-Viscous) is*

$$\mathbf{v}^\delta(t; \mathbf{x}) = -\frac{\delta}{2} \log \left\{ (2\pi\delta t)^{-n/2} \int_{\mathbb{R}^n} \exp\left(-\frac{1}{2\delta t} \cdot |\mathbf{x} - \mathbf{y}|^2\right) \exp\left(-\frac{2}{\delta} \cdot \mathbf{v}^\delta(\mathbf{y})\right) d\mathbf{y} \right\} \quad (5)$$

in $\mathbb{R}^n \times (0, T]$ and $\mathbf{v}^\delta(t; \mathbf{x}) = 0$ otherwise.

Corollary 1 (HJ solution as a log expectation). *On the domain of the differential equation, the unique solution can be written as the following expectation*

$$\mathbf{v}^\delta(t; \mathbf{x}) = -\frac{\delta}{2} \log \mathbb{E}_{\mathbf{y} \sim \mathcal{N}(\mathbf{x}, \delta t)} \left[\exp\left(-\frac{2}{\delta} \mathbf{v}^\delta(\mathbf{y})\right) \right]. \quad (6)$$

where $\mathbf{y}$ is drawn from a Gaussian distribution with mean $\mathbf{x}$ and variance δt .

Inspecting (Cole-Hopf), we may write

$$\mathbf{v}^\delta(t; \mathbf{x}) = -\frac{\delta}{2} \log(\varphi^\delta(t; \mathbf{x}))(\mathbf{x}) \text{ in } \mathbb{R}^n \times (0, T], \quad (7a)$$

from which we can write

$$\partial_{\mathbf{x}} \mathbf{v}^\delta(t; \mathbf{x}) = -\frac{\delta}{2} \cdot \frac{\nabla \varphi^\delta(t; \mathbf{x})}{\varphi^\delta(t; \mathbf{x})}. \quad (8)$$

Similarly, for $\mathbf{v}_t$, recall that $\varphi_t^\delta = \Omega'(\mathbf{v}^\delta) \mathbf{v}_t^\delta$, where $\Omega' = -(2/\delta) \exp(-\frac{2}{\delta} \mathbf{v}^\delta)$ is the derivative obtained from the linear transformation (A.4). It follows that

$$\mathbf{v}_t^\delta = \frac{\varphi_t^\delta}{\Omega'(\mathbf{v}^\delta)} := \frac{\frac{\delta}{2} \Delta \varphi^\delta}{-(2/\delta) \exp(-\frac{2}{\delta} \mathbf{v}^\delta)} \quad (9)$$

143 from which we can write the time derivative as

$$v_t^\delta = -\frac{\delta^2}{4} \frac{\Delta \varphi^\delta}{\exp(-(2/\delta)v^\delta)}. \quad (10)$$

144 We now cast the computation of the optimal “controllable”² reachable set into a form that leverages
 145 the Moreau envelope (Moreau-Env) in a successive approximation scheme so that the reachable set
 146 can be obtained via the subdifferential operator of the problem.

147 3.2 Operators and Envelopes of HJ Equations

148 Monotone operator theory has had a huge impact on partial differential equations and nonlinear
 149 systems (author?) (Zar71, BH14), reducing nonlinear analyses to problems of the form

$$\text{determine } x \in \text{zer } Q = \{x \in \mathcal{H} \mid 0 \in Qx\}. \quad (11)$$

150 The subdifferential operator (see (author?) (BCBC17, §16)), $\partial g : \mathcal{H} \rightarrow 2^{\mathcal{H}}$, of $g(x)$ at $\bar{x}$ is the set
 151 of all ϑ that satisfies

$$\{\vartheta \in \mathcal{H} \mid (\forall x \in \mathcal{H}) \langle x - \bar{x}, \vartheta \rangle + g(\bar{x}) \leq g(x)\}. \quad (12)$$

152 That is, g is subdifferentiable at x if $\partial g(x) \neq \emptyset$; hence, elements of $\partial g(x)$ are the subgradients of g
 153 at x .

154 An optimization implication of (12) is Fermat’s rule, which states that for $g : \mathcal{H} \rightarrow (0, T)$ (author?)
 155 (Mor65),

$$\text{argmin } g = \text{zer } \partial g. \quad (13)$$

156 The proximity operator of $g(x)$ at a time, $t \in (0, T]$ is the minimizers set (author?) (Mor62a)

$$\text{Prox}_{tg}(x) \triangleq \arg \min_{z \in \mathcal{H}} \left(g(z) + \frac{1}{2t} \|z - x\|^2 \right) \quad (\text{Prox-Op})$$

157 that define the Moreau envelope $v(\cdot, t)$, defined as (author?) (Mor62b)

$$v(x, t) = \min_{z \in \mathcal{H}} \left(g(z) + \frac{1}{2t} \|z - x\|^2 \right). \quad (\text{Moreau-Env})$$

158 The envelope v expands troughs of $g(x)$ and its local minimizers concur with those of g (author?)
 159 (COO⁺18).

160 **Lemma 2.** Suppose that $g(x)$ is BUC, its subdifferential exists, the set $\mathcal{S}_\alpha \triangleq \{z \in \mathbb{R}^n : g(z) \leq$
 161 $\inf g + \alpha\}$ is compact, $\text{zer } \partial g$ exists and $x \in \mathcal{S}_\alpha$. An important result established in (author?)
 162 (HWFO24, Lemma A.3) relates the spatial gradients of the solution to (HJ-IVP) to the proximity
 163 operator of $g(x)$ at time t as

$$\text{Prox}_{tg}(x) = x - t \partial_x v(t; x) \approx x - t \cdot \partial_x v^\delta(t; x). \quad (14)$$

164 3.3 HJ PDE’s spatial gradients and its proximity operator

165 Consider the kernel $\varphi^\delta(x, t)$ in (??) which corresponds to the solution of (HJ-Viscous). Resolving
 166 the convolution (??) is intractable for large n and time step t . However, it is same as the probability
 167 density of a Gaussian distribution with mean x and variance δt (as derived in (author?) (HWFO24,
 168 Appendix B)), i.e.

$$\partial \varphi_x^\delta(t; x) = -\frac{1}{\delta t} \cdot \mathbb{E}_{y \sim \mathcal{N}(x, \delta t)} [(x - y) \exp(g(y)/\delta)]. \quad (15)$$

169 Plugging (??) and (15) into (8), we find that

$$t \cdot \partial v_x^\delta(t; x) = x - \frac{\mathbb{E}_{y \sim \mathcal{N}(x, \delta t)} [y \exp(-g(y)/\delta)]}{\mathbb{E}_{y \sim \mathcal{N}(x, \delta t)} [\exp(-\delta^{-1} g(y))]} \quad (16)$$

²We use this term because the agents in the differential game system have control inputs

Algorithm 1 Sampling-based RCBRT descent.

Given $\{\epsilon, \delta\} > 0$, and initial step size α .
function **ProxReach**($\mathbf{x}, \mathbf{g}, \eta, t_0, T$) ▷ Sample size, η .
 Set $\tau = t_0, t_s = (T - t_0)/k$.
 while $T - \tau > \epsilon T$ **do**
 $\mathbf{y}[\tau] \leftarrow \mathcal{N}(\mathbf{x}, \delta)$; ▷ Draw finite samples from $\mathbf{x}$.
 $\mathbf{z}[\tau] \leftarrow \mathbf{g}(\mathbf{y}[\tau])$; ▷ See equation (7).
 $\partial_{\mathbf{z}} \mathbf{v}(\tau; \mathbf{z}) \leftarrow \text{Prox}_{\tau \mathbf{g}}(\mathbf{z}, \delta)$ ▷ Eq. (Prox-Op).
 $\tau \leftarrow \tau + t_s$
 $\mathbf{z}[\tau + t_s] \leftarrow \mathbf{z}[\tau] - \alpha \cdot \mathbf{g}[\tau]$;
 $\tau \leftarrow \text{Adjuster}(\tau, \mathbf{g}[\tau], \mathbf{g}[\tau - \Delta t])$;
 end while
end function

170 provides the time-scaled spatial gradients of $\mathbf{v}(t; \mathbf{x})$ in (19) without the need for unstable schemes
171 employed in upwinding methods within grid discretization numerical reachable sets' computational
172 algorithms (**author?**) (lev24, Mol25, Mit04).

173 Substituting the r.h.s of (16) into (14), we find that

Proximal Operator for $\partial_{\mathbf{x}} \mathbf{v}^\delta(t; \mathbf{x})$

$$\text{Prox}_{t\mathbf{g}}(\mathbf{x}) = \frac{\mathbb{E}_{\mathbf{y} \sim \mathcal{N}(\mathbf{x}, \delta t)} [\mathbf{y} \cdot \exp(-\mathbf{g}(\mathbf{y})/\delta)]}{\mathbb{E}_{\mathbf{y} \sim \mathcal{N}(\mathbf{x}, \delta t)} [\exp(-\delta^{-1} \mathbf{g}(\mathbf{y}))]} \quad (17)$$

174

175 3.4 Hamilton-Jacobi-Proximity Operator Relation

176 Now consider the HJI PDEs (HJI-RCBRT) and (HJI-RCBRAT) in light of (17). We can write the
177 domain of these PDEs in light of the proximal expression (17) as

$$\begin{aligned} \partial \mathbf{v}_t(t; \mathbf{x}) + \min\{0, \mathbf{H}(t; \mathbf{x}, \partial \mathbf{v}_\mathbf{x}(t; \mathbf{x}))\} &= 0, \\ \approx \partial \mathbf{v}_t^\delta(t; \mathbf{x}) + \min\{0, \mathbf{H}^\delta(t; \mathbf{x}, \partial \mathbf{v}_\mathbf{x}^\delta(t; \mathbf{x}))\} &= 0, \\ := \partial \mathbf{v}_t^\delta(t; \mathbf{x}) + \min \left\{ 0, \max_{\mathbf{u} \in \mathcal{U}} \min_{\mathbf{w} \in \mathcal{W}} \left\langle f(t; \mathbf{x}, \mathbf{u}, \mathbf{w}), \frac{1}{t}(\mathbf{x} - \text{Prox}_{t\mathbf{g}}(\mathbf{x})) \right\rangle \right\} &= 0, \end{aligned} \quad (18)$$

178 for (HJI-RCBRT). Similarly, the right-hand-side of the variational inequality for (HJI-RCBRAT) can
179 be written as

$$\begin{aligned} \min\{\partial \mathbf{v}_t(t; \mathbf{x}) + \mathbf{H}(t; \mathbf{x}, \partial \mathbf{v}_\mathbf{x}(t; \mathbf{x})), \mathbf{g}(\mathbf{x}, t) - \mathbf{v}(t; \mathbf{x})\}, \\ \approx \min\{\partial \mathbf{v}_t^\delta(t; \mathbf{x}) + \mathbf{H}^\delta(t; \mathbf{x}, \partial \mathbf{v}_\mathbf{x}^\delta(t; \mathbf{x})), \mathbf{g}(\mathbf{x}, t) - \mathbf{v}^\delta(t; \mathbf{x})\}, \\ := \min\{\partial \mathbf{v}_t^\delta(t; \mathbf{x}) + \max_{\mathbf{u} \in \mathcal{U}} \min_{\mathbf{w} \in \mathcal{W}} \langle f(t; \mathbf{x}, \mathbf{u}, \mathbf{w}), \partial \mathbf{v}_\mathbf{x}^\delta(t; \mathbf{x}) \rangle, \mathbf{g}(t; \mathbf{x}) - \mathbf{v}^\delta(t; \mathbf{x})\}. \end{aligned} \quad (19)$$

180 Thus, evaluating the spatial gradient of $\mathbf{v}^\delta$ in (19) amounts to drawing samples using the proximity
181 expression above and using (??) to evaluate $\mathbf{v}^\delta$. We can now write (HJI-RCBRAT) as

$$\min \left\{ \partial \mathbf{v}_t^\delta(t; \mathbf{x}) + \max_{\mathbf{u} \in \mathcal{U}} \min_{\mathbf{w} \in \mathcal{W}} \left\langle f(t; \mathbf{x}, \mathbf{u}, \mathbf{w}), \frac{1}{t}(\mathbf{x} - \text{Prox}_{t\mathbf{g}}(\mathbf{x})) \right\rangle, \right. \quad (20)$$

$$\left. \mathbf{g}(\mathbf{x}, t) + \delta \log \mathbb{E}_{\mathbf{y} \sim \mathcal{N}(\mathbf{x}, \delta t)} [\exp(-\delta^{-1} \mathbf{g}(\mathbf{y}))] \right\} \leq 0, \quad \mathbf{v}(0; \mathbf{x}) = \mathbf{g}(0; \mathbf{x}). \quad (21)$$

182 3.5 Problem Algorithmics

183 We now cast the solution of the variational HJI computational problem as a Monte Carlo algorithmics.
184 Towards finding global solutions to the optimization problem, we adapt the Hamilton-Jacobi Moreau

185 adaptive descent algorithm of (author?) (HWFO24) to the computation of the HJI(HJI-RCBRT) and
186 (HJI-RCBRAT).

187 The Moreau adaptive descent algorithm works by minimizing the Moreau envelope (Moreau-Env) of
188 the HJ PDE in a nonconvex optimization scheme via a zeroth-order sampling scheme. In its native
189 form, within a set convergence loop, it works by

- 190 • first computing the spatial gradient of $v^\delta(t; \mathbf{x})$; then
- 191 • updating the states of the dynamic system in a gradient descent scheme (under an appropriate
192 step size); and lastly
- 193 • uses an adaptive forward-backward time-update scheme to update the index of time.

194 The algorithm is succinctly described in Algorithm 1. Given a large initial step size α , time interval
195 $(t_0, T]$ and the state values for a given problem $\mathbf{x}$, the algorithm proceeds in a reactive sampling-based
196 gradient descent fashion by drawing finite samples of size η from the (potentially large) state space
197 $\mathbf{x} \in \mathbb{R}^n$. The samples on Line 5 of Algorithm 1 are draw from a multinomial Gaussian filter for
198 a total of η samples whose mean and variance are $\mathbf{x}$ and δt , respectively in every iteration of the
199 algorithm. Within line 7 of Algorithm 1, the Hamiltonian for the system is computed but negated
200 to make the computational scheme conform tot he backward reachable setting (since the iterative
201 loop proceeds forward in the index of time in Algorithm 1). In all our implementations, the time
202 index within the loop is adaptively adjusted (i.e. Line 9) based on a version of the Armijo-Goldstein
203 condition (author?) (Arm66) to ensure that an appropriate time step is taken in every iteration as we
204 seek the minimum value of the Moreau envelope.

205 After convergence, the solution(s) $\{\mathbf{x}\}$ that minimize the value function are returned. We now
206 adopt the HJ-MAD algorithm to the resolution of HJ problems with adversarial control inputs in a
207 two-player differential game. This is delineated in Algorithm 1. After the optimization of the extrema
208 of the value function, finding the safe set is equivalent to executing a search algorithm within the
209 optimized value function to evaluate where the condition (Zero Levelset) are fulfilled. In the next
210 section, we will connect the scheme we have presented so far to the safe reachability resolution
211 problem of two rockets on a cartesian grid as originally used in our previous paper (author?) (lev24).

212 3.6 Convergence analysis of the algorithm

213 TO-DO: Haoxiang, please follow the results in this link:

214 4 Numerical Results

215 In this section, we provide a representative problem and amend it to the viscous HJI
216 PDE (HJI-RCBRT) to test and verify our hypothesis.

217 4.1 The Rockets Launch Problem

218 We consider the rocket launch problem of Dreyfus (author?) (Dre66) which was amended to a
219 differential game between two identical rockets, $\mathbf{P}$ and $\mathbf{E}$, on an (x, z) cross-section of a Cartesian
220 plane in (author?) (lev24, Mol25). We want to compute the usable part (author?) (Isa99) or
221 RCBRT (author?) (MBT05) of the *approximate* terminal surface's boundary for a predefined target
222 set over a target tube. The usable part entails the regions of the state-space for which the min-max
223 operation over either strategy of $\mathbf{P}$ and $\mathbf{E}$ is below zero and where the boundary of the usable part
224 (BUP) constitute where (HJI-RCBRAT) is exactly zero.

225 For a two-player differential game, let $\mathbf{P}$ and $\mathbf{E}$ share identical dynamics in a general sense so that
226 we can freely choose the coordinates of $\mathbf{P}$; however, $\mathbf{E}$'s origin is a distance Φ away from (x, z) at
227 plane's origin (see Fig. 4.1) so that the $\mathbf{PE}$ vector's inclination measured counterclockwise from the
228 $\mathbf{x}$ axis is θ .

229 Hence, we may consider the current problem as a terminal pursuit-evasion *differential* game, where-
230 upon the game terminates when *capture* occurs. This eventual terminal represents the final *overap-*
231 *proximated* (non)-capture surface for the $\mathbf{P} - \mathbf{E}$ game (to be shortly introduced). We desire that

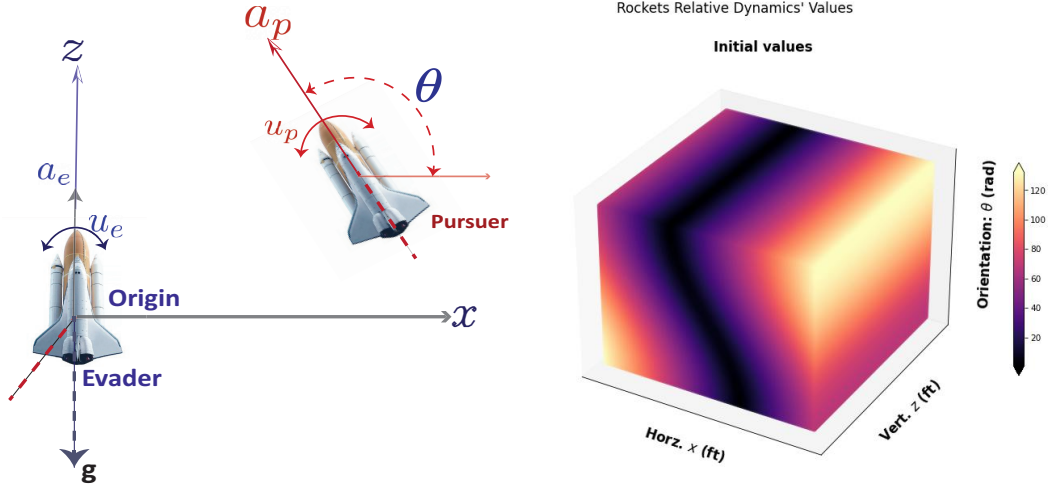


Figure 1: *Left*: Motion of two rockets on a Cartesian xz -plane with a thrust inclination in relative coordinates given by $\theta := u_p - u_e$. *Right*: Representation of the initial values as a heat map for the two rockets described in (27).

every $\max$ and $\min$ operations during a *game of kind* are in the interior of the plane where motion exists; this way, no sudden changes from extremes are too aggravating in cost.

Let the states of P and E be (x_p, x_e) , respectively, driven by thrusts (u_p, u_e) respectively (see Figure 4.1). Fix the rockets' range so that what is left of the motion of either P or E 's is restricted to orientation on the (x, z) plane as illustrated in Fig. 4.1. where a and g are respectively the acceleration and gravitational accelerations (in feet per square second)³. As long as E remains within this target region or *backward reachable tube* (or BRT), P cannot cause damage or exercise an action of deleterious consequence on, say, the territory being guarded by E . Setting up E to maximize a payoff quantity with the largest possible margin or at least frustrate the efforts of P with minimal collateral damage while the pursuer minimizes this quantity constitutes a terminal value *optimal* differential game: there is no optimal pursuit without an optimal evasion.

The motion of P relative to E 's along the (x, z) plane includes the relative orientation, the control input, shown in Fig. 4.1 as $\theta = u_p - u_e$. Following the conventions in Fig. 4.1, the game's relative equations of motion in *reduced space* (author?) (Isa99, §2.2) was derived as (see (author?) (lev24, §VI.A)) is $\mathbf{x} = (x, z, \theta)$ where $\theta \in [-\frac{\pi}{2}, \frac{\pi}{2}]$ and $(x, z) \in \mathbb{R}^2$ are

$$\dot{\mathbf{x}} \implies \{\dot{x} = a_p \cos \theta + u_e x, \dot{z} = a_p \sin \theta + a_e + u_e x - g, \dot{\theta} = u_p - u_e.\} \quad (22)$$

The capture radius of the origin-centered circle Φ (we set $g = 1.5$ ft) is $\|PE\|_2$ so that $g^2 = x^2 + z^2$. All capture points are specified by the variational HJ PDE (author?) (MBT05):

$$\frac{\partial v}{\partial t}(\mathbf{x}, t) + \min \left[0, H\left(\mathbf{x}, \frac{\partial g(\mathbf{x}, t)}{\partial \mathbf{x}}\right) \right] \leq 0, \quad (23)$$

with Hamiltonian given by

$$H(\mathbf{x}, p) = - \max_{u_e \in [\underline{u}_e, \bar{u}_e]} \min_{u_p \in [\underline{u}_p, \bar{u}_p]} [p_1 \quad p_2 \quad p_3] \begin{bmatrix} a_p \cos \theta + u_e x \\ a_p \sin \theta + a_e + u_p x - g \\ u_p - u_e \end{bmatrix}. \quad (24)$$

Here, p are the co-states, and $[\underline{u}_e, \bar{u}_e]$ denotes extremals that the evader must choose as input in response to the extremal controls that the pursuer plays i.e. $[\underline{u}_p, \bar{u}_p]$. Rather than resort to analytical *geometric reasoning*, we may analyze possibilities of behavior by either agent via a principled numerical simulation. This is the essence of this work. From (24), set $\underline{u}_e = \underline{u}_p = \underline{u} \triangleq -1$ and

³We set $a = 1 \text{ ft/sec}^2$ and $g = 32 \text{ ft/sec}^2$ in our simulation.

254 $\bar{u}_p = \bar{u}_e = \bar{u} \triangleq +1$ so that $\mathbf{H}(\mathbf{x}, p)$ is

$$\begin{aligned}
& - \max_{u_e \in [\underline{u}_e, \bar{u}_e]} \min_{u_p \in [\underline{u}_p, \bar{u}_p]} \begin{bmatrix} p_1(a_p \cos \theta + u_e x) + \\ p_2(a_p \sin \theta + a_e + \\ u_p x - g) + p_3(u_p - u_e) \end{bmatrix} \\
& \triangleq -ap_1 \cos \theta - p_2(g - a - a \sin \theta) - \bar{u}|p_1 x + p_3| + \underline{u}|p_2 x + p_3|, \quad (25)
\end{aligned}$$

255 where the last line follows from setting $a_e = a_p \triangleq a$. Thus, the rockets HJI equation (23) becomes

$$\partial_t \mathbf{v}(t; \mathbf{x}) + \min [0, (-ap_1 \cos \theta - p_2(g - a - a \sin \theta) - |p_1 x + p_3| - |p_2 x + p_3|)] \leq 0, \quad (26)$$

256 where p_1, p_2, p_3 are the coefficients of the co-states obtained from (19). A similar derivation can be
257 obtained for the (HJI-RCBRAT). The initial values of the game is as illustrated in ??.

258 4.2 Numerical Integration Considerations

259 We take the infinitesimal physics problem above and make it algorithmic first by discretizing the
260 spatial and temporal domains i.e.

$$(x, z) \in (-100, +100], \theta \in (-\pi/2, \pi/2], t \in (0, 1], \quad (27)$$

261 with 1000 points along the temporal dimension and $\mathbf{x}_k - \mathbf{x}_{k-1} = 1/100$ along the spatial dimensions,
262 where $k \in \mathbb{N}$. All our codes are implemented in pytorch, that runs on an Azure Linux node with the
263 cpu processor specification: Intel Xeon Silver 4208 at 2.1GHz and 65GB memory; The pytorch runs
264 on a Ubuntu 22.04 operating system.

265 We defined the target set as the top-left corner of the state space ?? and then run the Algorithm 1 on a
266 time range of $(0, 1]$ seconds; we chose Dirichlet boundary conditions

$$\mathbf{v}^\delta(0; \mathbf{x}) = \mathbf{g}(0; \mathbf{x}) \triangleq 0. \quad (28)$$

267 The spatial gradients (co-states) were computed using the transformation (17) which were then used
268 in generating the Hamiltonian and subsequently the solution to the safe set using (23).

269 5 Conclude

270 We have presented a gradient-based nonconvex optimization scheme for computing the safe set of a
271 two-person differential game. The gradient scheme is a zero order Moreau envelope optimization
272 scheme whose extrema coincides with that of the HJ-Isaacs equation (which is non-convex in its
273 Hamiltonian). We provided a means to optimize the Proximal operator of the Moreau envelope
274 by adapting the (author?) (HWFO24) algorithm and we note that once this extrema is found,
275 determining the safe set amounts to searching within the resulting optimized terminal values of the
276 HJI equation to see where the conditions of (Zero Levelset) are fulfilled. We demonstrated a proof of
277 concept with the relative motion of two rockets on a Cartesian grid where both players have control
278 inputs: the evading player must avoid the pursuing player within a pre-specified time range while the
279 pursuing player must try to capture the evader within the given time by maximizing time. In a best
280 response strategy, each player responds to the control input of its adversarial player in a best-response
281 strategy. Our results demonstrate the feasibility of our approach.

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A Lemmas and Proofs

In this section, we construct the machinery we use in building our results.

A.1 Unique Solution to the viscous HJ Equation (HJ-Viscous)

Let us quickly construct the solution to the viscous initial-value-HJ problem.

Proof of Lemma 1. Recall from (HJ-Viscous) that

$$\mathbf{v}_t^\delta - \frac{\delta}{2} \Delta \mathbf{v}^\delta + \mathbf{H}(t; \mathbf{x}, \partial_x \mathbf{v}^\delta) = 0 \text{ in } \mathbb{R}^n \times (0, T], \quad \mathbf{v}^\delta(0; \mathbf{x}) = \mathbf{g}(\mathbf{x}) \text{ on } \mathbb{R}^n \times \{t = 0\}. \quad (\text{A.1})$$

Let $\varphi^\delta(t; \mathbf{x}) = \Omega(\mathbf{v}^\delta(t; \mathbf{x}))$, where $\Omega : \mathbb{R}^n \rightarrow \mathbb{R}$ is a smooth function. Our goal is to choose Ω so that φ^δ solves a linear partial differential equation. We want Differentiating, we find that $\varphi_t^\delta = \Omega'(\mathbf{v}^\delta) \mathbf{v}_t^\delta$ and $\Delta \varphi^\delta = \Omega(\mathbf{v}^\delta) \Delta \mathbf{v}^\delta + \Omega''(\mathbf{v}^\delta) \mathbf{H}(t; \mathbf{x} \partial_x \mathbf{v}^\delta)$. Thus, (HJ-Viscous) implies

$$\varphi_t^\delta = \Omega' \mathbf{v}_t^\delta \triangleq \Omega'(\mathbf{v}^\delta) \left(\frac{\delta}{2} \Delta \mathbf{v}^\delta - \mathbf{H}(t; \mathbf{x}, \partial_x \mathbf{v}^\delta) \right) \quad (\text{A.2a})$$

$$= \frac{\delta}{2} \left(\Delta \varphi^\delta - \Omega''(\mathbf{v}^\delta) \mathbf{H}(t; \mathbf{x} \partial_x \mathbf{v}^\delta) \right) - \Omega'(\mathbf{v}^\delta) \mathbf{H}(t; \mathbf{x}, \partial_x \mathbf{v}^\delta) \quad (\text{A.2b})$$

$$= \frac{\delta}{2} \Delta \varphi^\delta - \left(\frac{\delta}{2} \Omega''(\mathbf{v}^\delta) + \Omega'(\mathbf{v}^\delta) \right) \mathbf{H}(t; \mathbf{x}, \partial_x \mathbf{v}^\delta). \quad (\text{A.2c})$$

We thus have

$$\varphi_t^\delta \triangleq \frac{\delta}{2} \Delta \varphi^\delta \quad (\text{A.3})$$

if we set the differential equation $\frac{\delta}{2} \Omega''(\mathbf{v}^\delta) + \Omega'(\mathbf{v}^\delta)$ to 0 — this is possible if we choose

$$\Omega(\mathbf{v}^\delta) = \exp \left(-\frac{2\lambda}{\delta} \right). \quad (\text{A.4})$$

Whence, if $\mathbf{v}^\delta$ solves the partial differential equation (HJ-Viscous), we must have (Cole-Hopf) i.e.

$$\varphi^\delta = \exp(-2\mathbf{v}^\delta/\delta). \quad (\text{A.5})$$

Equation (Cole-Hopf) is the so-called Cole-Hopf transformation (**author?**) (Eva22, §4.4.1); it serves as the solution to the initial-value problem for the heat equation (with conductivity $\delta/2$),

$$\begin{aligned} \partial_t \varphi^\delta(t; \mathbf{x}) - \frac{\delta}{2} \Delta \varphi^\delta(t; \mathbf{x}) &= 0 & \text{in } \mathbb{R}^n \times (0, T] \\ \varphi^\delta(t; \mathbf{x}) &= \exp(-2\mathbf{v}^\delta/\delta) & \text{on } \mathbb{R}^n \times \{t = 0\}. \end{aligned} \quad (\text{Heat-Equation})$$

Whence, the unique bounded solution to (Cole-Hopf) is

$$\varphi^\delta(t; \mathbf{x}) = (2\pi\delta t)^{-(n/2)} \int_{\mathbb{R}^n} \exp \left(-\frac{1}{2\delta t} |\mathbf{x} - \mathbf{y}|^2 \right) \exp \left(-\frac{2}{\delta} \mathbf{v}^\delta(\mathbf{y}) \right) d\mathbf{y}, \quad \mathbf{x} \in \mathbb{R}^n, t > 0. \quad (\text{A.6})$$

And since we can write $\mathbf{v}^\delta = (\delta/2) \log \varphi^\delta$, it follows that the unique bounded solution to the initial-value (HJ-Viscous) is (5), i.e.

$$\mathbf{v}^\delta(t; \mathbf{x}) = -\frac{\delta}{2} \log \left\{ (2\pi\delta t)^{-n/2} \int_{\mathbb{R}^n} \exp \left(-\frac{1}{2\delta t} |\mathbf{x} - \mathbf{y}|^2 \right) \exp \left(-\frac{2}{\delta} \mathbf{v}^\delta(\mathbf{y}) \right) d\mathbf{y} \right\} \quad (\text{A.7})$$

□

364 *Proof of Corollary 1.* The proof is straightforward. Examining the solution (5), and reordering
365 (Cole-Hopf), it is obvious that the right-hand-side is a form of the multivariate Gaussian distri-
366 bution if we set the mean to x and the variance to δt . *A fortiori*, the solution to the HJ equation can
367 be obtained by sampling from the multivariate Gaussian distribution (6). $\square$

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